Querying a Codebase with NVIDIA Auto Ontology

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A source-to-SQL adapter, demonstrated on Auto Ontology's own repository

Abstract

Can a database agent answer questions about software if the source is represented as tables? We built **code-ontology-bridge**, a Python extractor that turns source files, symbols, calls and existing docstrings into relational data and a semantic model for NVIDIA Auto Ontology. A second export, the semantics layer, adds verified views for calls, callbacks, reachability, file and symbol facts, dependencies and inheritance. We tested both on 48 questions about four repositories, written by separate AI authors who never saw what the bridge adds, re-derived by independent annotators and frozen before any model ran. Each answer counts as correct only if its SQL also returns the right rows on eight counterfactual versions of the database. Asked directly, DeepSeek v4.1 Flash rose from 20.6 correct answers with the schema alone to 34.0 with the exported model, and NVIDIA Nemotron 3.5 Lightning from 2.4 to 9.4 with the semantics layer, averaged over five runs. Inside Auto Ontology's own agent, the semantics layer was the only import that raised the score above Auto Ontology's own catalog of the database, by a small margin on 12 questions. The questions come from AI authors rather than the repositories' maintainers, and the extractor misses some calls that the source makes.

1. The question and the contribution

Suppose a developer changes `resolve_semantic_fks`, the function that resolves semantic foreign keys in Auto Ontology. Which callers reach it? Searching for its name finds an immediate caller. Reaching the scheduler also requires following a wrapper and recognizing a function passed as an argument. The question spans several files and two kinds of relationship: direct calls and callback references.

A source map can record those relationships once and make them available to SQL. NVIDIA Auto Ontology already has the other half of this workflow: an agent that retrieves semantic context, writes SQL and executes it against a connected database. This project connects the two. Instead of asking Auto Ontology about a sales database, we ask it about a database of its own implementation.

The work has three parts:

- **A source adapter.** A standalone extractor records Python source facts and exports DuckDB rows, model YAML and an evaluation dataset using Auto Ontology's existing contracts.
- **An exercised integration.** A compatibility patch configures public model clients and exposes an existing SQL routing option at the chat boundary. The retained run connects ingestion, model import, semantic compilation and question answering.
- **A code semantics layer.** Six verified views that define direct calls, callback references, reachability, file and symbol facts, dependencies and inheritance once, documented in the native model.
- **A blind, repeated study.** 48 questions about four repositories, written without sight of the bridge, scored on real and counterfactual databases, with every run, response and failure published.

The source-map approach comes from **CodeMap**, Will Cook's earlier source-navigation tool.

This bridge adapts that approach into an installable package; it does not depend on CodeMap's private interface. NVIDIA supplies the database agent and semantic layer. The bridge supplies the source-derived data and its model.

2. How the source reaches the agent

Auto Ontology needs both a live database to query and semantic context that explains the database. The bridge exports them separately. Figure 1 follows the two paths to SQL execution.

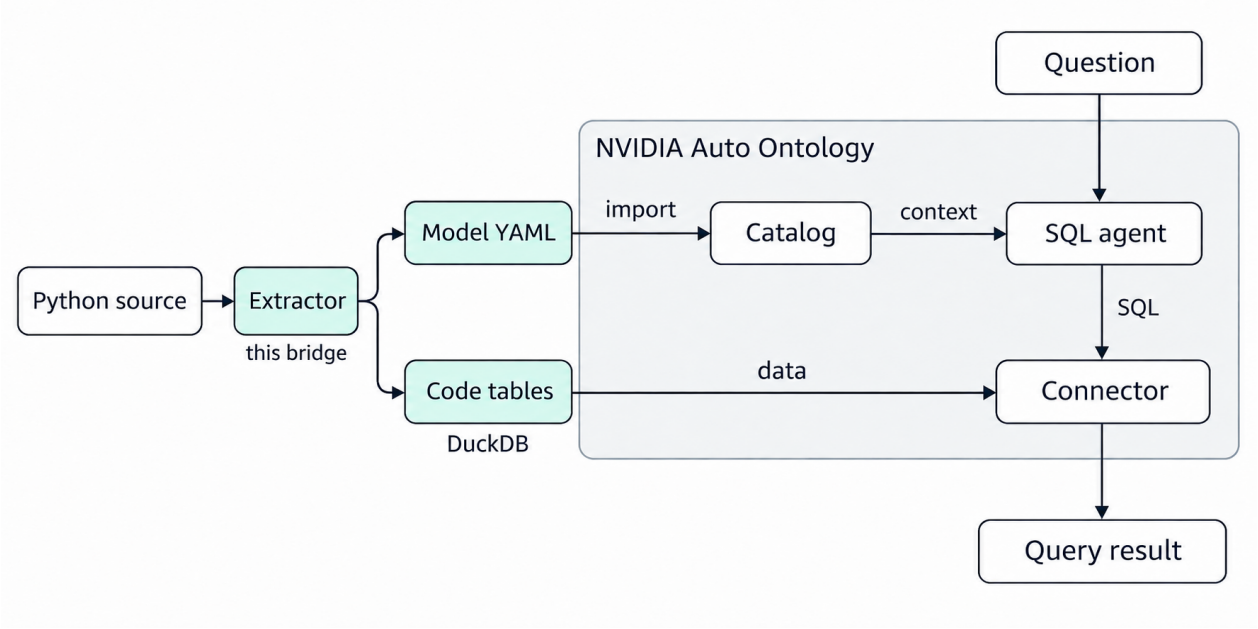


Figure 1: Two exports serve different purposes. DuckDB supplies source-code rows; model YAML supplies descriptions and relationships through import and compilation. The outlined components belong to Auto Ontology. The extractor is this bridge; the catalog is the semantic catalog and the connector is the existing DuckDB connector. This simplified query path omits ingestion services, model providers and answer narration.

The **DuckDB export** contains eight tables: files, docstring headings, symbols, class bases, calls, imports, test links and a source snapshot. Two views summarize incoming calls and public symbols without test associations. Auto Ontology reads these through its existing DuckDB connector.

The **model YAML** describes tables, terms, attributes and joins. It declares keys that the pinned DuckDB connector does not report. The live runner first ingests the database, maps the exported catalog identifiers to the server's identifiers, imports the model and compiles the semantic layer. Importing YAML alone does not create a queryable database.

A third export supplies DDL, CSV data and reference questions in NVIDIA's evaluation format. It provides a route toward a controlled comparison, but the full upstream benchmark harness was not run for this study.

What is a row, and what is a definition?

The extractor uses Python's AST and preserves source paths and line ranges. A function becomes a row in `code.symbols`. Its docstring becomes a value in `summary`. A call becomes a row in `code.calls`, with identifiers for its caller and, when resolved, its callee. Joining symbols to files lets a query recover locations or exclude test files.

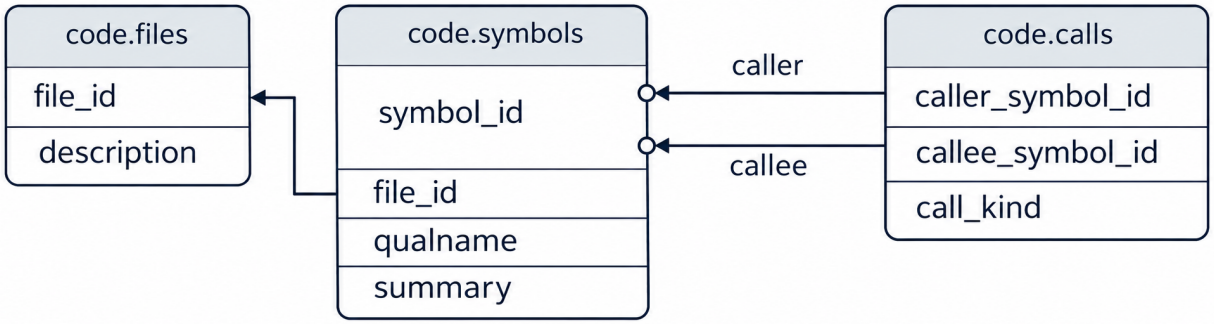


Figure 2: Selected columns show the joins used by a caller query. Each call references its caller and, when resolved, its callee; each symbol references its file. Descriptions of particular files and functions remain data values. The semantic model describes what the tables and columns mean.

This distinction keeps the model small: it defines a symbol as an entity type, rather than creating a separate semantic term for every function. Source descriptions are copied from existing docstrings. Missing descriptions stay empty; the extractor does not ask a model to invent them. The interface specification gives all columns, identity rules and resolution categories.

One source occurrence, two representations

In `semantic_scheduler.py` under `auto_ontology/ingestion_service/`, line 203 of the pinned source passes the wrapper as an argument:

```
tables_processed = await asyncio.to_thread(
    run_semantic_compilation, database_name
)
```

The extracted row includes these fields:

```
call_kind: callback    resolution: imported
confidence: low       line: 203
```

The caller is `SemanticScheduler._run_once`, whose definition starts at line 132 in the scheduler file. The callee is `run_semantic_compilation`, defined in `compile.py` under `auto_ontology/semantic/`.

These selected fields from the frozen native model input show how Code symbol maps to the data:

```
id: cob:auto_ontology_code:term:symbols
name: Code symbol
represents:
  - cob:auto_ontology_code:table:code.symbols
columns_attributes:
  - id: cob:auto_ontology_code:attribute:code.symbols.symbol_id
    name: symbol id
    column_id: cob:auto_ontology_code:column:code.symbols.symbol_id
```

Below, IDs omit the shared `cob:auto_ontology_code:` prefix. The Call site term represents `table:code.calls`. Its semantic foreign keys connect two columns to the symbol-ID attribute:

- `column:code.calls.caller_symbol_id`
- `column:code.calls.callee_symbol_id`

Both target `attribute:code.symbols.symbol_id`. These definitions describe the joins; the particular functions and call occurrences remain database rows. The `summary` attribute's authored

description explains a source-docstring column, while each row's `summary` contains the original docstring. Both the queryable rows and the imported semantic context are needed. This is Auto Ontology's native format, not Ossie YAML.

The runner remaps exported catalog IDs before import. The frozen exported input and a later `/api/terms` response are retained; the submitted YAML, ID mapping and full compiled-model export are not. The terms response is a partial catalog readback and does not show which definitions a question retrieved. The inspection backend exposes these stages separately and supplies hash-checked source excerpts at build time.

What changed upstream?

The data connector, model import and SQL pipeline already existed. The source extractor and exports need no changes to NVIDIA's code or model credentials. The recorded live run used Auto Ontology at `cdaf61e1bd0cf3820d11d72b43bb460816aebbd3`; compatibility with later upstream revisions has not been established. A separate twelve-file compatibility patch configures public model clients, forwards the existing `calculation_only` option through the chat API and extends a placeholder validator. It adds no code-specific connector, SQL engine or query planner.

Fixed SQL routing matters here because a question such as "What does this module do?" can sound like a metadata question. In this dataset, the answer is a value in a table. The patch selects the row-querying path for every question in the run. Its documentation and validation bind the patch to 611 passing tests in the upstream query, chat and utility scopes. That is scoped validation, not the full upstream test suite or evidence of upstream acceptance.

3. Following the caller chain

Question 11 asks which resolved caller symbols reach `resolve_semantic_fks`, directly or indirectly, excluding tests and reporting the shortest depth. This is a summary of the request: the full input also prescribed column names and types and instructed the agent to traverse resolved calls backwards, recursively, including callback references. Given those instructions, Auto Ontology generated recursive SQL over `code.calls` and returned four callers.

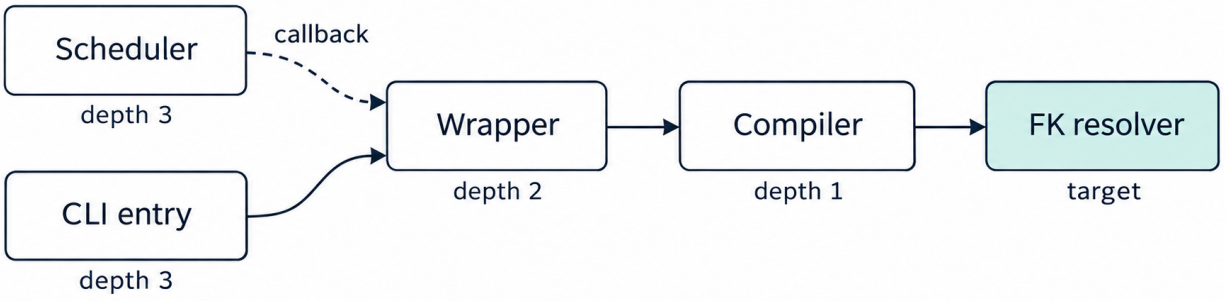


Figure 3: Four callers reach the foreign-key resolver through a wrapper and compiler. The table below identifies each source symbol. Arrows point from caller to callee, and SQL follows them in reverse. Solid edges are direct calls. The dashed edge records a callback passed to asyncio.to_thread, not proof that it ran.

Diagram label	Source symbol	Depth
Compiler	`_run_semantic_compilation`	1
Wrapper	`run_semantic_compilation`	2
Scheduler	`SemanticScheduler._run_once`	3
CLI entry	`main`	3

The target labeled "FK resolver" is `resolve_semantic_fks`; it is not a returned caller.

The locally computed inspection presets return three callers with direct calls alone and four when callback references are included; only the scheduler is added. This is a traversal of saved rows, not another model answer.

The scheduler passes `run_semantic_compilation` to `asyncio.to_thread`. The bridge records that argument as a callback reference, which connects the scheduler to the same chain as the ordinary caller `main`. The generated query follows reverse edges, avoids cycles and groups by full caller identity to select the minimum depth.

The saved response includes the request, intermediate events, final SQL and complete symbol identifiers. The request contains a hint about reverse traversal and callbacks; it is not an unaided discovery test. No saved SQL example was imported for this run. The answer establishes reachability within the extracted map. Dynamic calls can be missing, and a callback reference does not establish execution.

4. Evaluation: questions the system never saw

4.1 Design

The study (protocol) asks how much each part of what the bridge exports contributes to correct answers, on questions written without knowledge of it. The order of work is fixed by public commits.

1. **The semantics layer came first.** Six DuckDB views over the exported tables define common code-question concepts once: `call_edges` (direct calls and callback references for each caller and callee), `reachability` (the shortest chain of calls between two symbols, through direct calls only and through callbacks too, unbounded and safe on cycles), `symbol_profile`, `file_profile` (including each file's folder), `file_dependencies` and `class_ancestry`. Every view and column is documented in the native model with keys and joins. Property tests check each view against an independent Python computation on random code maps. The layer and its tests were committed before any question existed.
2. **Four authors wrote 12 questions each**, one per repository: NVIDIA Auto Ontology at a current upstream commit (474 Python files), pallets/click (91), encode/httpx (60) and Textualize/rich (213). The authors were separate AI agents with the source checkout and a database containing only the base tables, instructed not to open the layer, its tests or the study folder. They chose topics freely, wrote reference SQL and checked each answer against the source.
3. **A second annotator per repository** received only the question text and output columns and wrote its own SQL. All 48 answers agreed with the authors'. Wherever the annotator named a second plausible reading, the question was reworded so one reading remains: 17 questions, with no reference changed and none removed.
4. **The questions, references and protocol were committed and pushed** before any model saw them.

4.2 Arms and scoring

Four briefings of the same model, querying the same database:

- **A, schema.** Table and column names and types.
- **B, base model.** A plus every table's and column's description, the keys and the joins: the model this project has always exported.
- **V, views.** A plus the semantics layer's views listed by name and type, without descriptions; a control for the views alone.
- **S, semantics layer.** B plus the layer's views, documented columns, keys, joins and a short glossary of definitions.

Inside Auto Ontology, the agent with its retrieval, validation and repair, the database contents and the chat model are fixed, and only the import changes: **nothing** (Auto Ontology's own catalog of the database), the **base model**, or the **semantics layer**.

DeepSeek v4.1 Flash and NVIDIA Nemotron 3.5 Lightning answered through OpenRouter, at temperature 0 with reasoning off. Arms A, B, V and S ran five complete times; the Auto Ontology arms three times each on the 12 Auto Ontology questions.

Each answer is scored twice. **Exact rows**: the returned rows equal the reference rows, with named and typed columns and the requested order. **Robust**: the same SQL also returns the reference SQL's rows on eight counterfactual copies of the repository's database, which add calls, callbacks, same-named functions, methods, imports and bases, remove some calls and change docstrings and sizes. A query can return the right rows on one database by coincidence; following Zhong, Yu and Klein (2020), the counterfactual databases expose many such queries. The reference answer changes on at least one counterfactual database for 44 of the 48 questions. Substituting the reference SQL for the model scores 48 of 48 on both measures.

4.3 Results

A model asked directly, robust answers out of 48, mean of five runs (lowest and highest in brackets):

Briefing	DeepSeek v4.1 Flash	Nemotron 3.5 Lightning
A, schema	20.6 (18–23)	2.4 (2–4)
B, base model	34.0 (32–37)	5.6 (5–6)
V, views without descriptions	28.0 (26–31)	2.8 (2–5)
S, semantics layer	33.8 (31–37)	9.4 (7–12)

Inside Auto Ontology, robust answers out of 12, mean of three runs:

Auto Ontology imports	DeepSeek v4.1 Flash	Nemotron 3.5 Lightning
Nothing	7.3	1.3
Base model	7.3	2.0
Semantics layer	8.7	2.7

The exported models make a large difference to a model asked directly. With DeepSeek, the base model adds 13 robust answers out of 48; question by question, 14 improve and one regresses. With Nemotron the semantics layer lifts robust answers from 2.4 to 9.4; eight questions improve and none regress.

The layer's value depends on the model and the agent. For DeepSeek asked directly, the base model and the semantics layer tie (six questions improve and six regress). For Nemotron the layer is clearly better than the base model. The views without their descriptions do worse than the base model for both, so the documentation carries the effect.

Inside Auto Ontology, the semantics layer gives the highest score for both models. For DeepSeek, Auto Ontology's own catalog of the database scores the same as the imported base

model; for Nemotron the base model raises the robust score from 1.3 to 2.0 and the layer to 2.7. For DeepSeek the layer, whose views encode reachability, callbacks and folders, raises the robust score from 7.3 to 8.7. The difference is small in absolute terms, on 12 questions and three runs.

4.4 Limits

The questions were written by AI authors under the protocol's rules, not by the repositories' maintainers, and their topics reflect what the base database can answer. Reference answers are correct for the extracted database. The authors' source checks found facts the extractor misses: calls through a local alias, calls to functions with `@overload` stubs, names reached through package re-exports and calls on objects of unknown type. Each is recorded with its question. Nemotron scores low on these questions throughout with reasoning off. The Auto Ontology arms used revision `cdaf61e` with a twelve-file compatibility patch and took a median of three to four minutes per question; arms A to S took one to three seconds and cost under a dollar in total.

An earlier experiment with 38 author-written questions about Auto Ontology alone, one run per arm, found the same direction for the base model and shaped this design; its records are kept with the earlier protocol and results.

5. What the source map records

Before the earlier experiment, two extractor defects were fixed and its database rebuilt: imports now bind only in the scope that contains them, and calls in a nested function count for its enclosing function only when that function calls or passes on the nested one. The rebuild removed 21 of 9,442 linked calls and added 7; the question 11 answer did not change.

The map itself is incomplete. In that earlier snapshot of Auto Ontology, of 31,178 direct call sites, 9,442 resolve to an internal callee and 11,794 are classified as dynamic attribute calls without one. Another 1,046 module-level calls are omitted from the call table. The small residual category named `unresolved` does not measure all missing edges. Callback records have lower confidence; test links reflect imports or matching filenames rather than measured coverage. Repeated qualified definitions, such as property getters and setters, receive distinct line-qualified identities, while ambiguous targets remain unresolved.

Queryable source representations are established techniques. CodeQL builds databases for source analysis, and CodexGraph lets language-model agents query code graph databases. The contribution here is the adapter to Auto Ontology's contracts and its demonstrated use on source-derived rows.

6. Reviewing and extending the integration

For an Auto Ontology maintainer, the work separates into three reviewable pieces:

Review question	Implementation or evidence
How do source facts enter the existing interfaces?	Extractor, exporters and interface specification.
What did the live service need?	Compatibility patch, with configuration, boundaries and scoped tests.
What did the agent actually produce?	Detailed results, linking every run, response and score.

The immediate maintainer question is whether software repositories are a useful semantic-catalogue use case, and how public-model configuration and SQL-routing requirements should be supported.

The next tests are questions written by the repositories' maintainers, a comparison against an agent reading the source directly, and the semantics layer against a current Auto Ontology revision.

Separate governance commands record certification history, annotate SQL and reconcile definitions. They do not enforce certified-only retrieval or prove which definitions the agent used. This evaluation makes no new claim about those commands.

The runs used Auto Ontology `cdaf61e` and bridge `2d3686c`; the two DeepSeek Auto Ontology runs started at the next commit, `e1d41b3`, which changed only the website and the results summariser. Each Auto Ontology run's manifest binds full revisions, model settings, the database digest and frozen inputs. Follow the reproduction guide for extraction, compatibility tests or a new live run. Figure notes document the illustrations' sources and omissions.